Loc Nguyen

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SUMMARY

Data Scientist with demonstrated experience in SQL, Python and Power BI, building automated data pipelines and interactive dashboards that translate complex datasets into performance insights for senior stakeholders. Delivers measurable analytical outcomes across structured environments, including a 20% reduction in data misclassification rates through rigorous data auditing and governance, and production model deployments achieving 94% mean precision supported by end-to-end Apache Airflow pipelines. Brings a genuine interest and curiosity in supporting analytical processes and developing end-to-end pipelines.

EXPERIENCE

ONETASK — **Melbourne, VIC**
Data Analytics Intern — July 2025 - November 2025

- Developed and benchmarked computer vision models (**YOLOv4, YOLO-NAS,** and **RF-DETR**) for traffic violation detection on Linux systems, producing proof-of-concept evaluations that informed model selection and business recommendations for future production deployments.
- Operationalised detection models enhanced with the Red-Fox Optimisation algorithm, achieving **94% mAP** for seatbelt detection and **82% mAP** for phone-use detection, moving solutions from experimentation to reliable, real-world traffic safety applications.
- Created an interactive **Power BI dashboard** translating trade-offs across **20 experiment settings** of computer vision models into performance visualisations, enabling clear strategic recommendations for decision-makers.
- Built an **Apache Airflow** pipeline to automate end-to-end training, testing, inference, and visualisation workflows, reducing manual intervention by **30%** and improving reproducibility across analytical cycles.

CONSTELLATION TECHNOLOGIES LIMITED — **Melbourne, VIC**
Data Analytics Intern — July 2024 - November 2024

- Developed a **BERT-based NLP question-answering system** to extract structured data from unstructured food product records, achieving **95%** extraction accuracy and improving the speed and consistency of data retrieval.
- Built an **Isolation Forest** anomaly detection model to identify abnormal temperature patterns with **99% accuracy,** supporting earlier detection of quality risks in agricultural supply chains and maintaining products in excellent condition.
- Continuously improved **Power BI dashboards** by implementing threshold alert features and driver analysis based on SHAP values, delivering actionable operational insights to business managers and supporting data-informed service decisions.

ONETASK — **Melbourne, VIC**
Data Evaluation Specialist — March 2023 - April 2026

- Reviewed up to **3,500** traffic enforcement images daily, producing true-positive and false-negative statistics that directly supported **model evaluation** and road-safety performance reporting.
- Designed and maintained **Power BI visualisations** of offence trends by highway and camera location, communicating high-volume hotspot patterns of **300+** daily offences to public-sector stakeholders and supporting enforcement prioritisation.
- Conducted monthly Excel data quality audits (VLOOKUP, Pivot tables) to validate outcomes against verified records, reducing the **misclassification rate** by **20%** and improving labelled dataset reliability to be used for strengthening model accuracy.
- Identified systemic image quality issues (blur, poor lighting) and escalated to engineering teams, enabling faster camera recalibration and preventing downstream data loss.

EDUCATION

Monash University **Clayton, Australia**
Bachelor of Applied Data Analytics Advanced (Honours) December 2025
First Class Honours (H1) - Dean's List 2023
Honours Thesis: Application of Detection Transformers in Live Traffic Monitoring Systems

PROJECTS

EUROPEAN BANK CHURN ANALYSIS

- Wrote **SQL** queries to extract data and build **RFM segmentation** logic from a 10,000-customer dataset; cleansed and transformed inputs via **Power Query**, then built **Power BI dashboards** with **DAX** used to create flags of high-churn risk factors (geography, age group, tenure), surfacing retention risks and tailoring retention recommendations per customer segment.
- Managed experiment tracking with **MLflow** and hyperparameter optimisation with **Optuna**, selecting the best **XGBoost** model at **72%** accuracy and the model was deployed via **Streamlit** for real-time churn predictions.

ROSSMANN FORECASTING PROJECT (Monash Kaggle Competition)

- Led a five-member team through full data lifecycle delivery by arranging weekly meetings for planning ahead and resolving technical issues, pre-processing and feature engineering in Python (Pandas), applying K-means customer sales segmentation (**98% accuracy**) and Random Forest forecasting (**95% accuracy** in 4-week sales prediction).

TECHNICAL SKILLS

Languages: SQL, Python (Pandas, Numpy, Matplotlib, Scikit-Learn, Pytorch, Jupyter), R, Excel (VLOOKUP, Pivot tables)
Visualisation: Power BI (DAX, Power Query), Matplotlib
Data & Pipelines: Apache Airflow, MLflow, Optuna, OpenCV, Docker, Streamlit
Machine Learning: Linear Regression, Logistic Regression, Decision Trees, Random Forest, KNN, K-means, Anomaly Detection (Isolation Forest), Regularisation Techniques (Ridge, Lasso)
Infrastructure: Git, Linux (Ubuntu), Docker
Core Competencies: Investment Analytics Support, Performance & Risk Reporting, Data Quality Auditing, Governance & Metadata Documentation, Full Data Lifecycle Management, Stakeholder Communication, Continuous Improvement, Predictive Modelling